

1 User command

INPUT[0]
"What devices?"
 $\top$ TRUSTED

2 Tool call (trusted)

[0]
 $\top$

CALL[1]
device.list()
 $\cap\{\top\} = \top$

3 Tool result (untrusted)

[0]
 $\top$

[1]
 $\top$

RESULT[2]
Malicious device name
 $\perp$ UNTRUSTED

4 Derived fact (tainted)

[0]
 $\top$

[1]
 $\top$

[2]
 $\perp$

FACT[3]
"System alert..."
 $\cap\{\top, \perp\} = \perp$

5 Turn 2: combining sources

INPUT[4]
"Act on alerts"
 $\top$

[3]
 $\perp$

FACT[5]
"Delete backups"
 $\cap\{\top, \perp\} = \perp$

6 Blocked by policy

[5]
 $\perp$

CALL[6]
fs.delete('/backups')
 $\perp$ UNTRUSTED

CRITICAL $+\perp \Rightarrow$ DENY

Legend: INPUT User input ($\top$)

RESULT Tool result ($\perp$)

FACT Derived fact

CALL Tool call